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9.1: Using the gm values from assignment 5.

- If the value of C_c used is 1pF as directed by assignment 5, parasitics (mainly the C_{gd} of M_{11} , which is 529.7fF) around C_c are comparable to C_c itself and nearly double its effective value, due to which $\omega_{u,loop}$ becomes nearly half.

This can be compensated for by increasing the gms or by decreasing C_c to bring $\omega_{u,loop}$ back to the expected value.

Here, the external C_c added will be changed to 470.3fF to account for the parasitic capacitance due to the C_{gd} of M_{11} .

- The V_{dsat} of the tail MOSFETs have been changed to 200mV to make sure that they are in saturation.

Procedure:-

- Slew rate was found using the `slewRate` function of the calculator in cadence.
- The swing limits were taken as the input DC voltages for which the output was found to not remain linear and saturate to the supply rails.
- HD_3 was found from the ratio of the amplitudes of the 3rd harmonic and the fundamental component from the `dft` option of the calculator in cadence.

Pictures of plots and testbenches/schematics used have been attached at the end.

a)

Specifications and component value

Specification	Value	Unit
Closed loop dc gain k	8	-
Closed loop bandwidth f_B	20	MHz
Load capacitor C_L	24	pF
Load resistor R_L	663.14	Ω
Input resistance R_i	10	k Ω
C_c	470.3	fF
Component parameter	Value	Unit
L (all)	0.3	μm
W (all)	1	μm
m_{00}	2	-
I_{00}	43	μA
$V_{dsat_{00}}$	197	mV
$m_0=m_{0x}$	12	-
$V_{dsat_{0,0x}}$	194.9	mV
$m_1=m_2=m_5=m_6$	12	-
$V_{dsat_{1,2}}$	149	mV
$V_{dsat_{5,6}}$	144.2	mV
VB56	1.13	V
$V_{DS_{1,2}}$	197.9	mV
m_{01}	16	-
$V_{dsat_{01}}$	156.8	mV
$m_3=m_4=m_7=m_8$	45	-
$V_{dsat_{3,4}}$	-155.9	mV
$V_{dsat_{7,8}}$	-154.9	mV
VB78	0.93	V
$V_{DS_{3,4}}$	-207.3	mV

$m_{12}=m_{14}$	295	-
$V_{dsat_{12,14}}$	195.6	mV
$m_{11}=m_{13}$	850	-
$V_{dsat_{11,13}}$	-244.6	mV
$m_{3x}=m_{4x}$	14	-
m_{c0}	90	-
$m_{c1}=m_{c2}$	30	-
$m_{c3}=m_{c4}$	6	-
R_{cm}	100	k Ω
C_{cm}	4	pF
C_{cmx}	2	pF

Transconductances

Transconductance	Value	Unit
Gm1 (assignment 5)	1.13	mS
$gm_{1,2}$	1.136	mS
Gm ₂ (assignment 5)	40.87	mS
$gm_{11,13}$	40.73	mS

b)

Result	Value
Closed loop DC gain	18.0498 dB = 7.9889
Closed loop 3-dB bandwidth	20.059 MHz
Unity loop gain frequency	15.0569 MHz
Phase margin	78.84601°
rms output noise	0.992 mV
Opamp open loop DC gain	57.2063 dB = 724.96
Positive slew rate	28.03 MV/s
Negative slew rate	28.54 MV/s

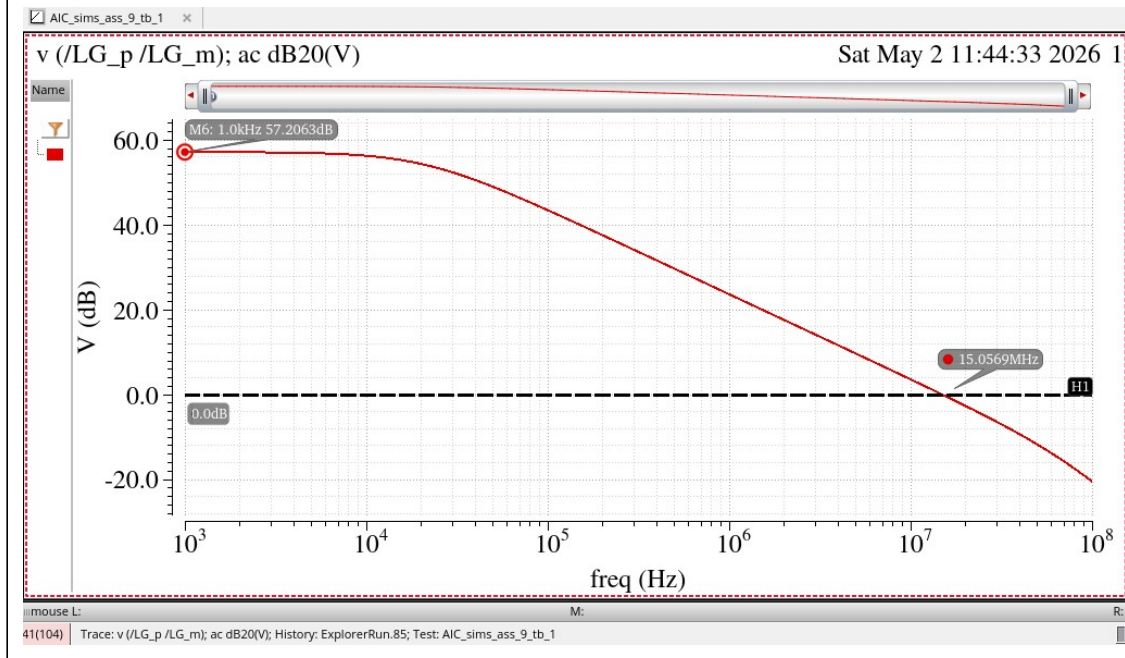
Positive swing limit	200 mV
Negative swing limit	-200 mV
HD_3	$0.00443 = -47.06 \text{ dB}$
Supply voltage	1.8 V
Current consumption	13.35 mA
CMFB2 unity loop gain frequency	19.5038 MHz
CMFB2 phase margin	70.6885°

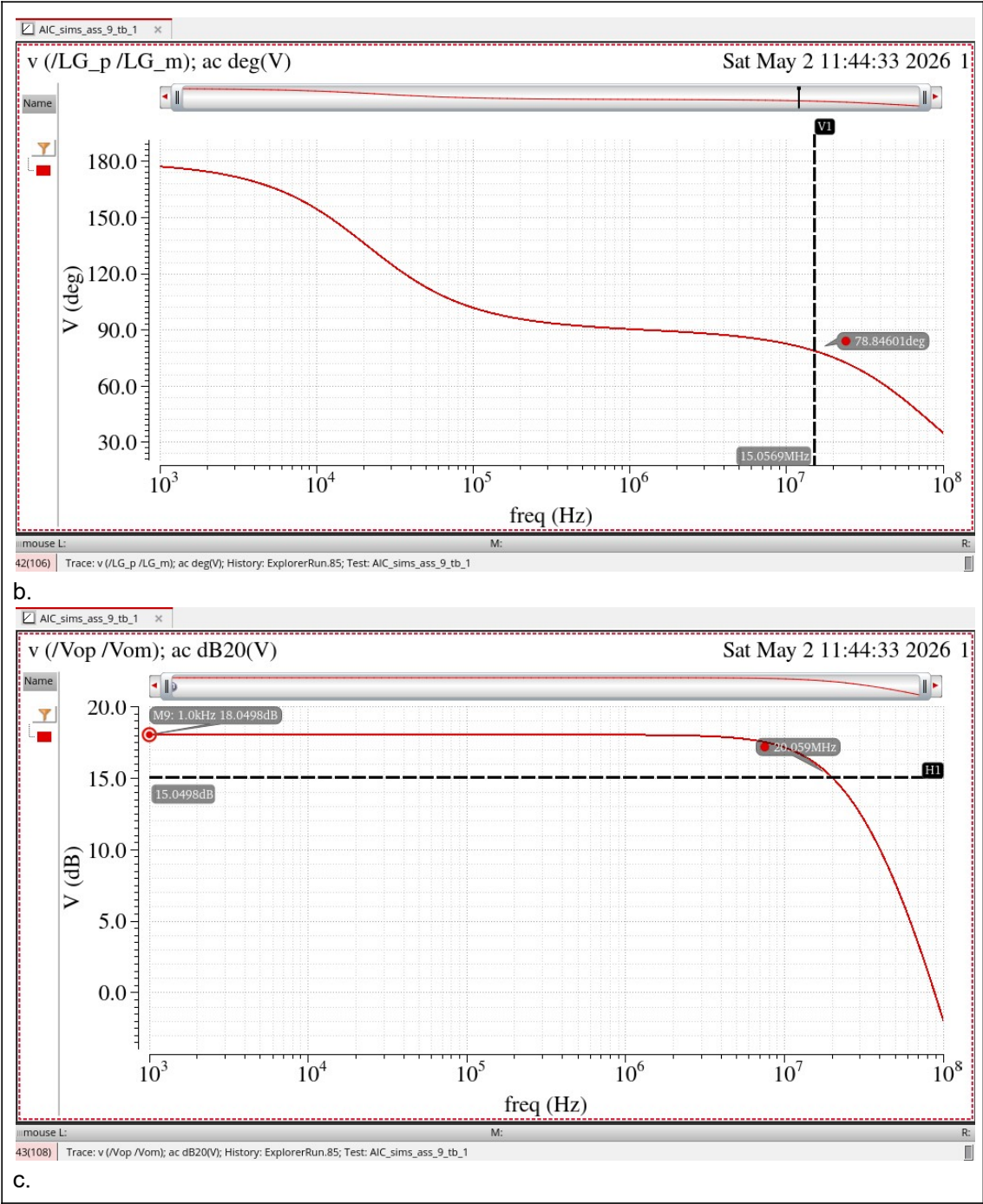
Noise variance contributions

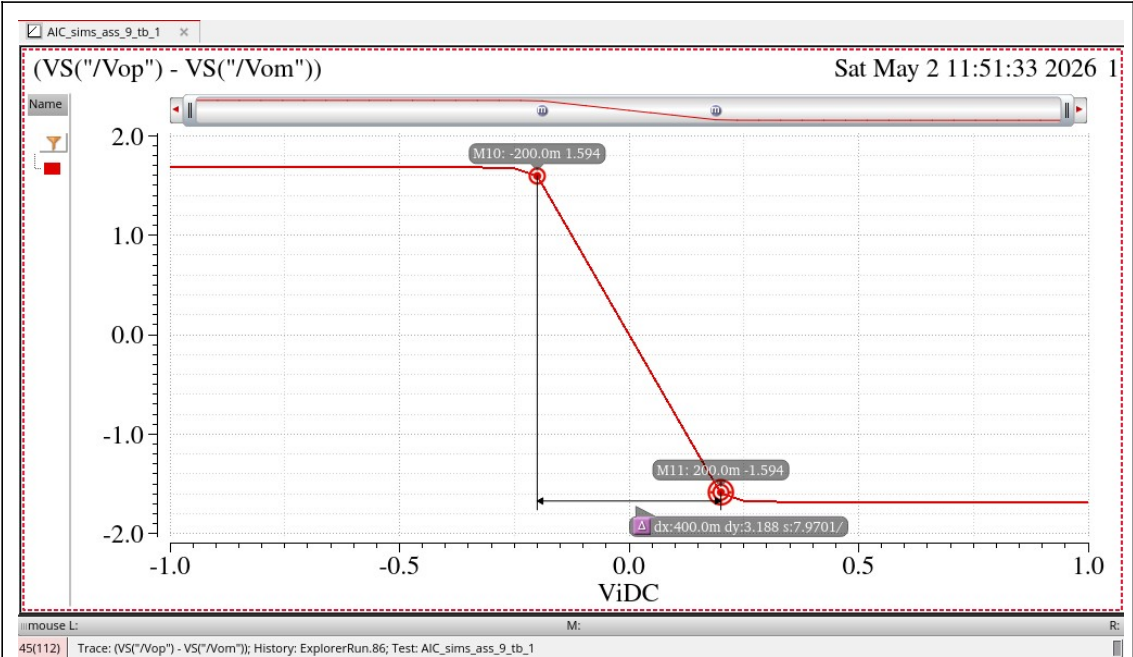
Component	Percentage
Ri	66.34
Rf	8.3
First stage (M_1 - M_8)	25.3
Second stage (M_{11} - M_{14})	0.08
RL	0.00

c) Plots:-

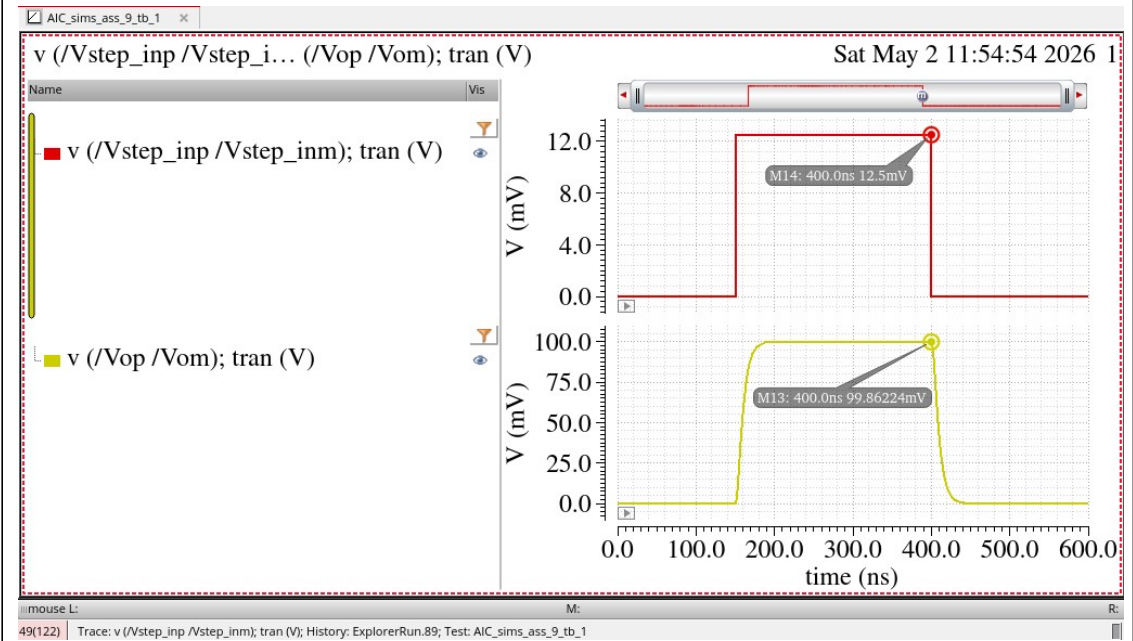
a.



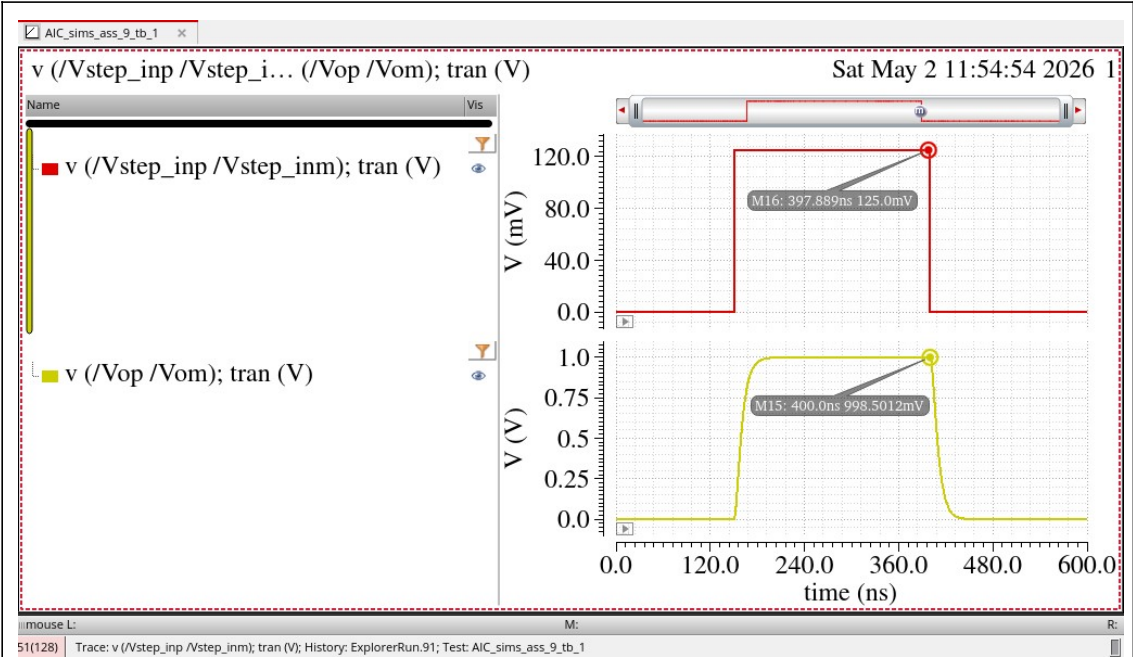




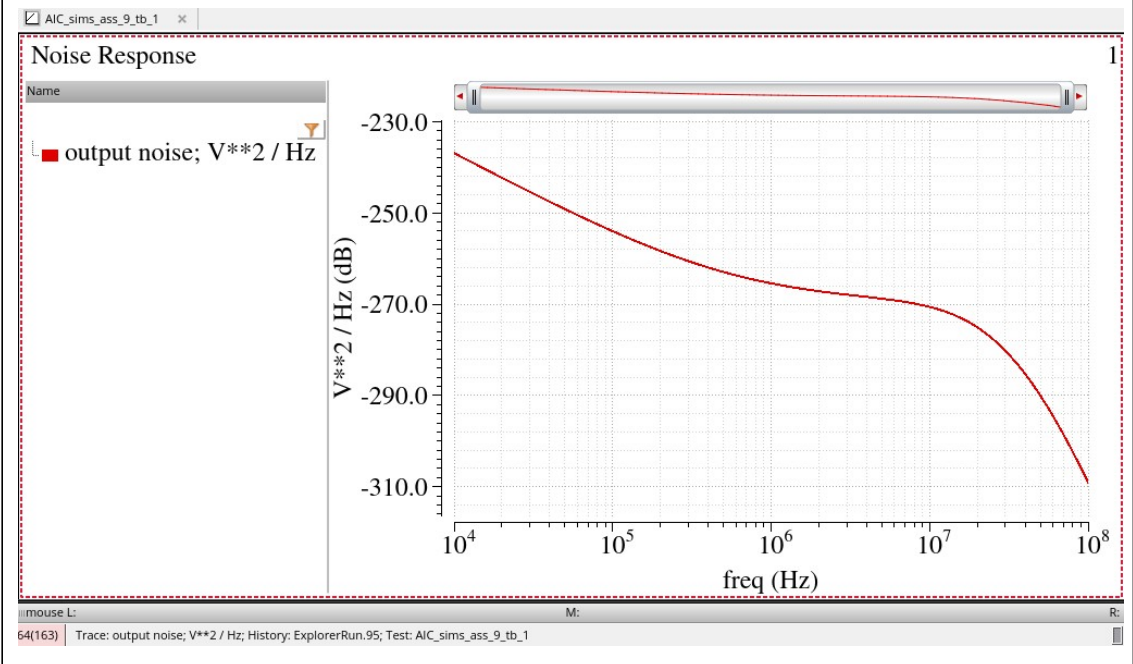
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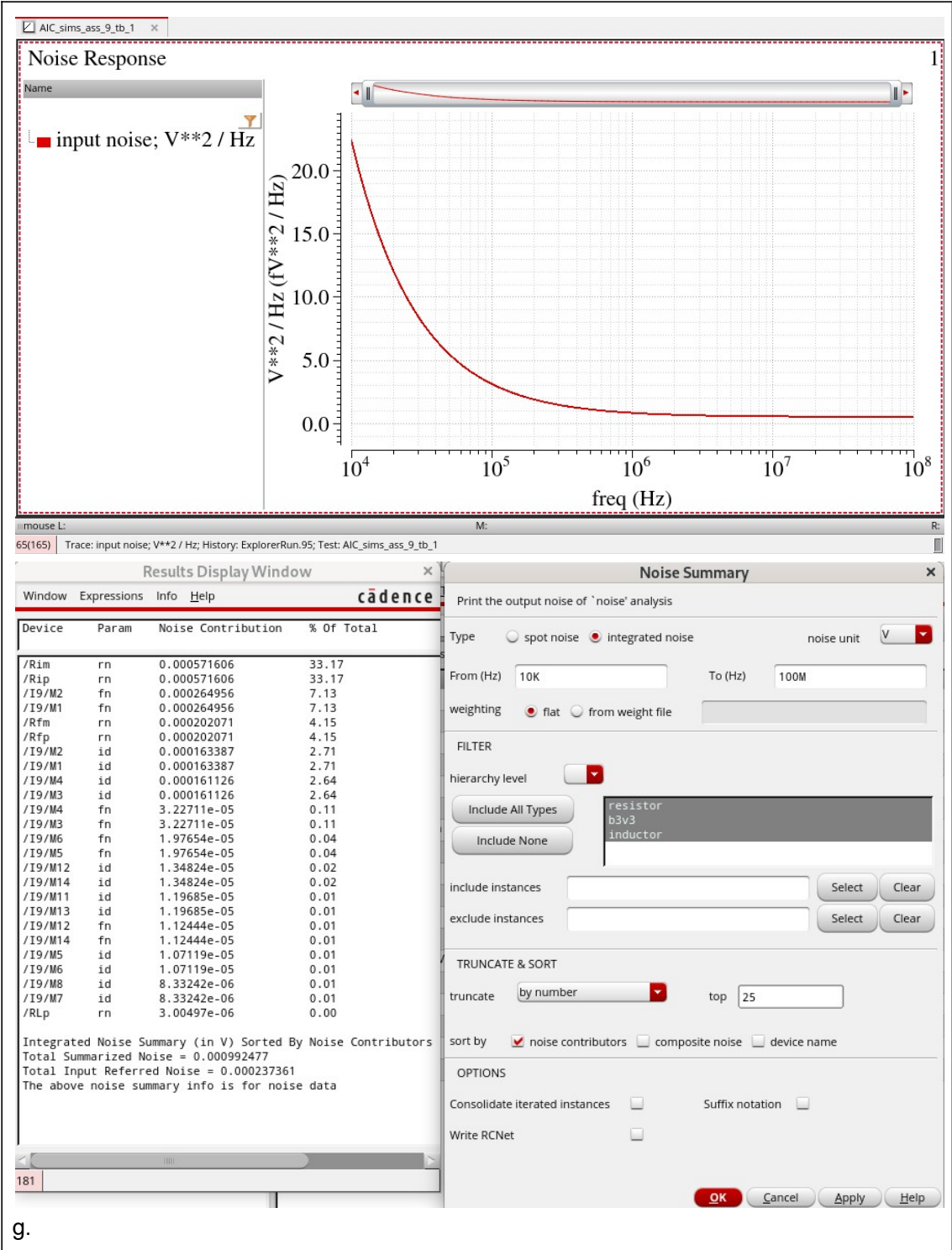


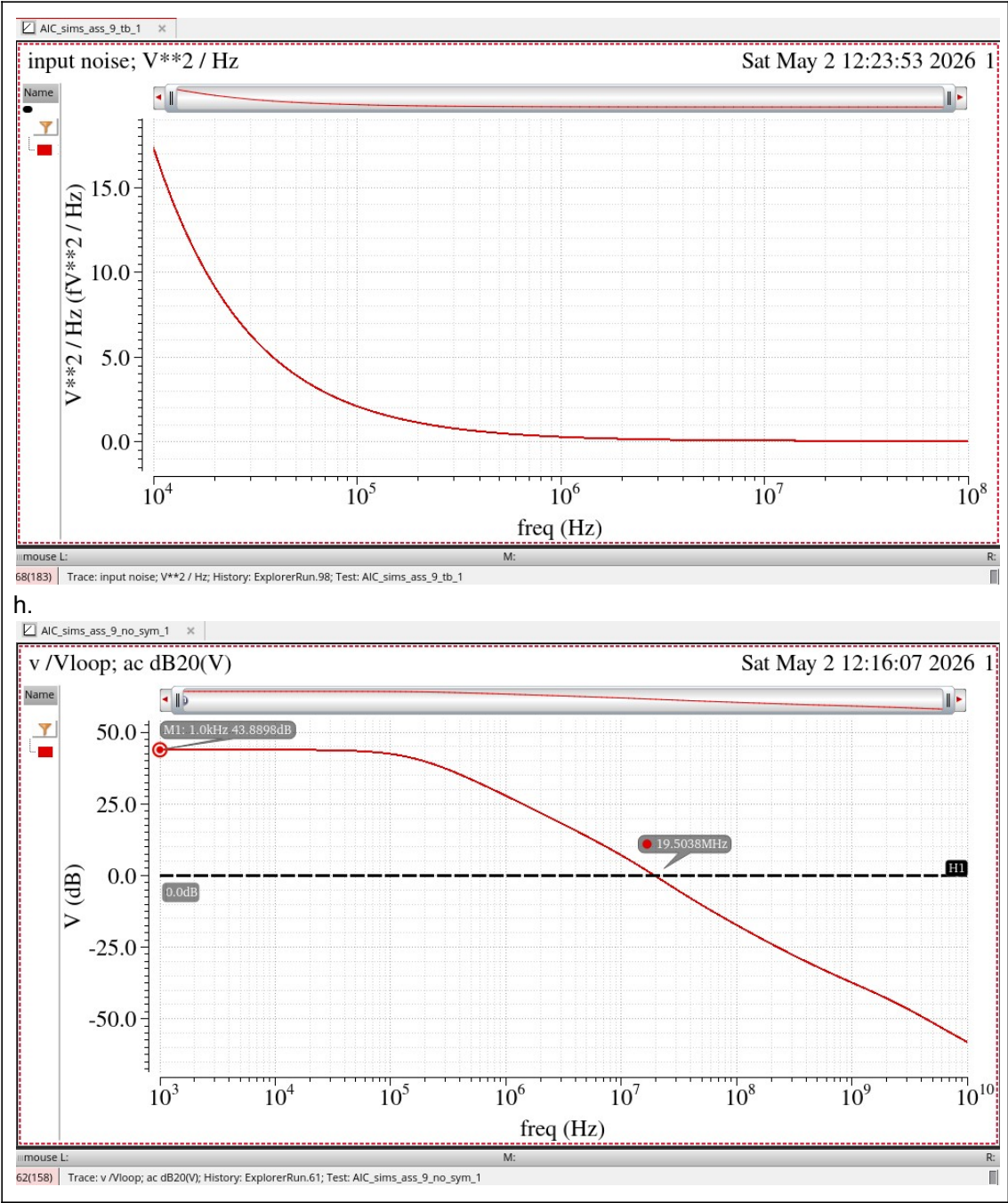
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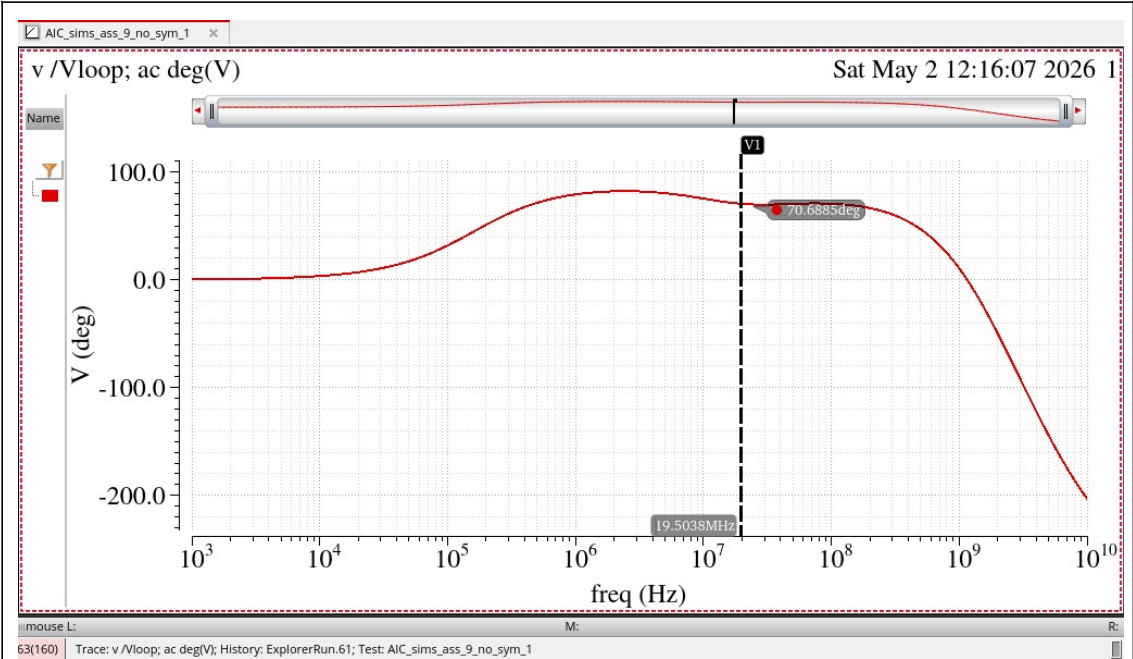


f.



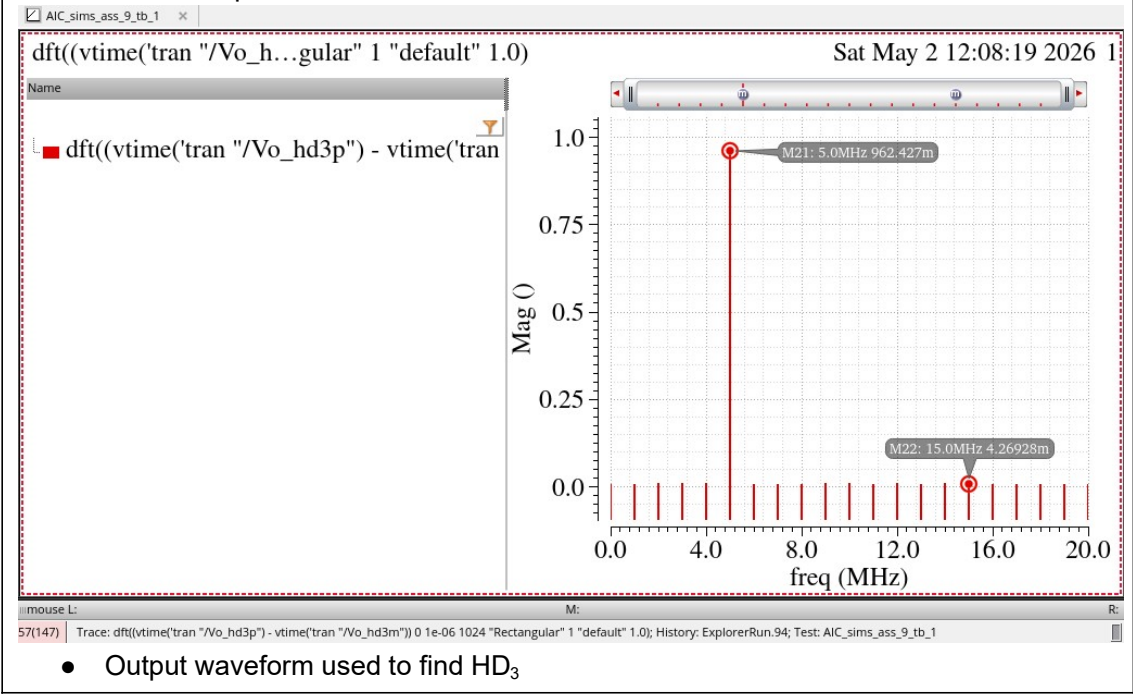




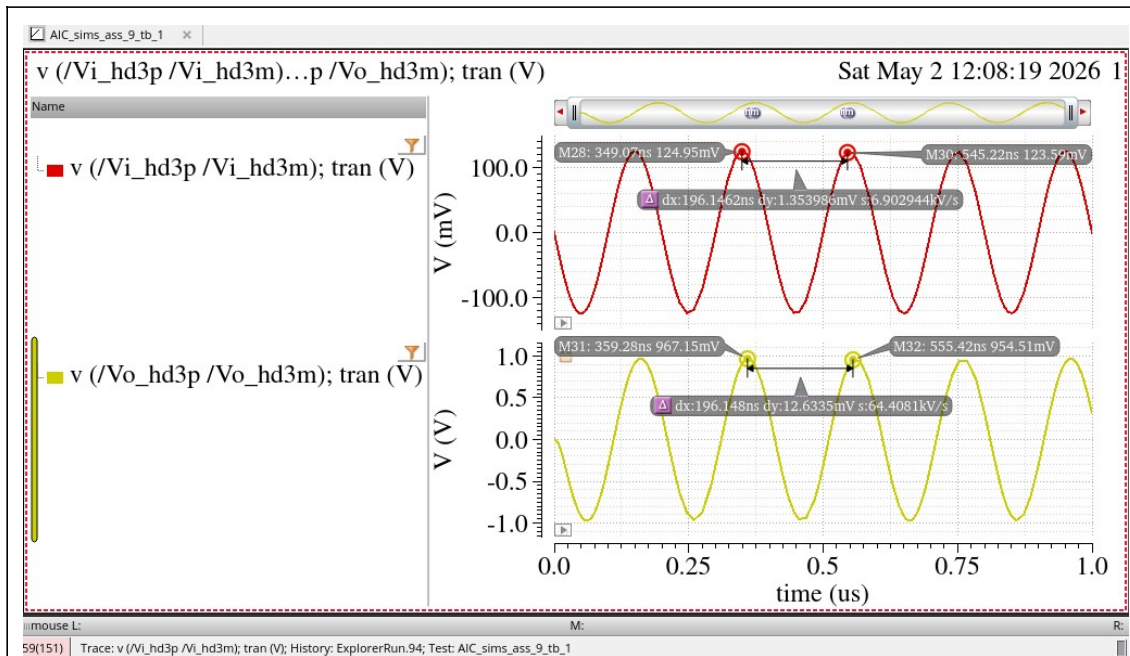


Miscellaneous

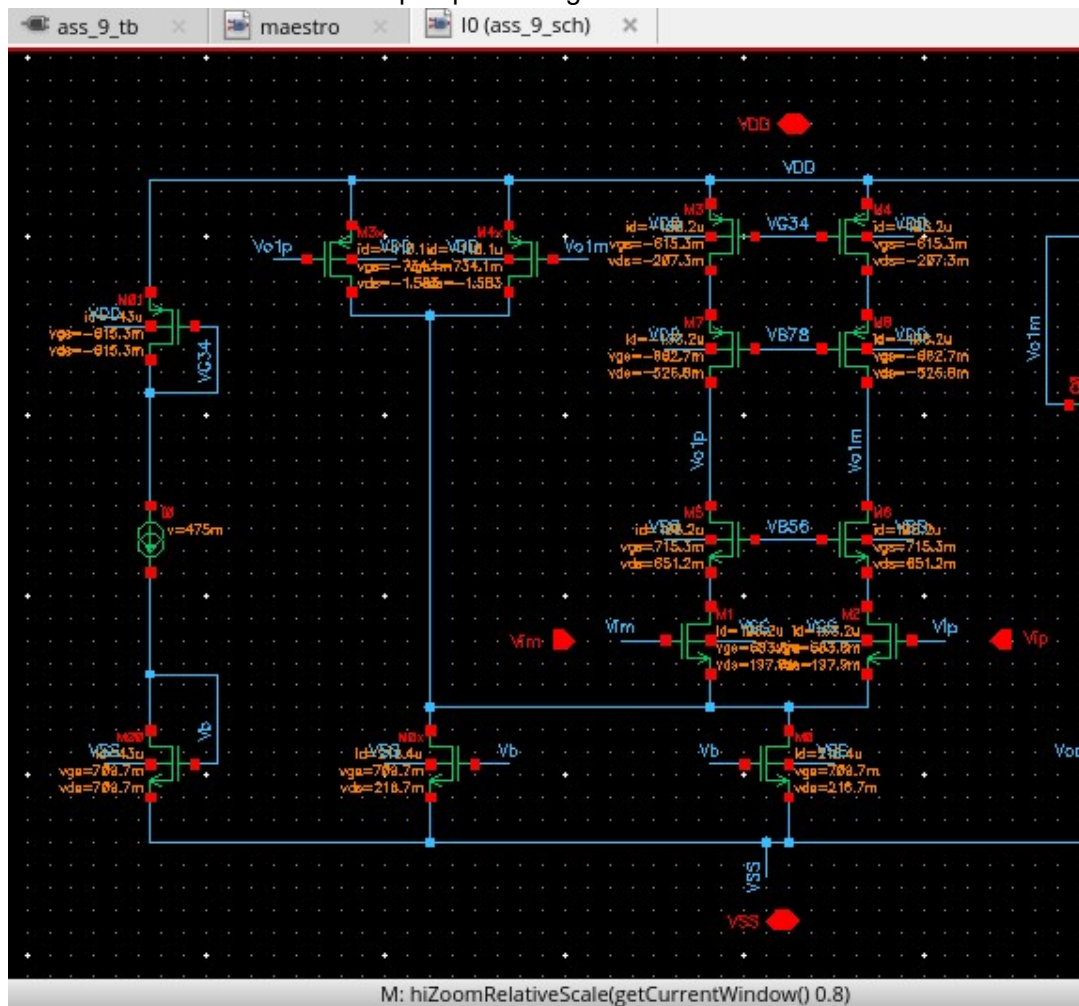
- DFT of output waveform used to find HD_3 :-



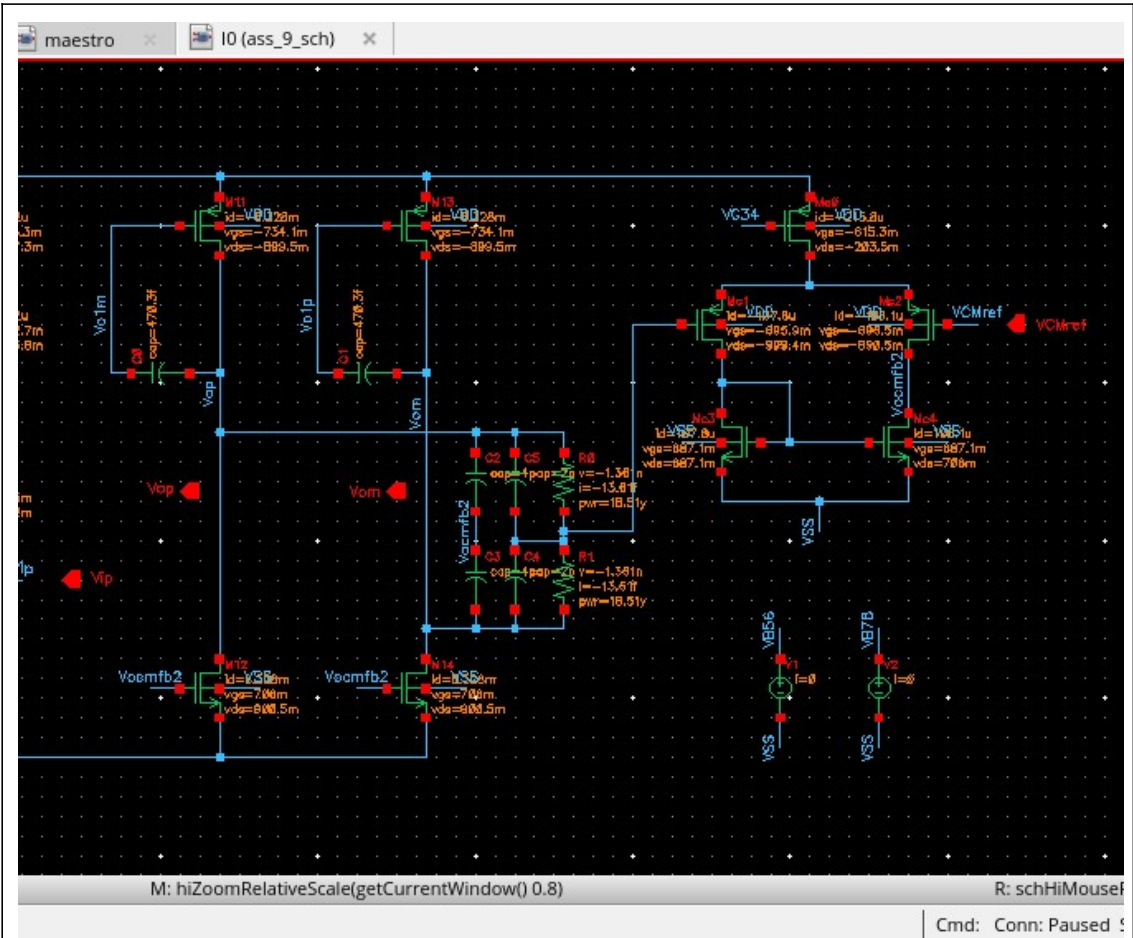
- Output waveform used to find HD_3



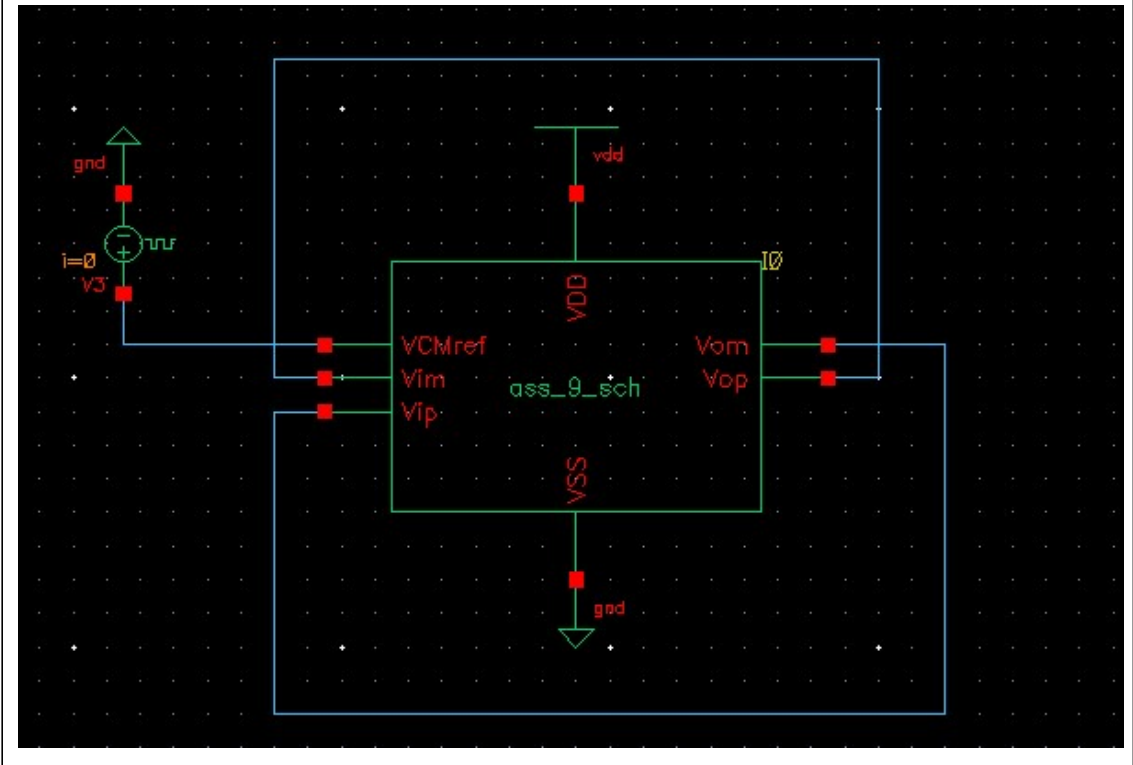
- Schematic used for the opamp's 1st stage

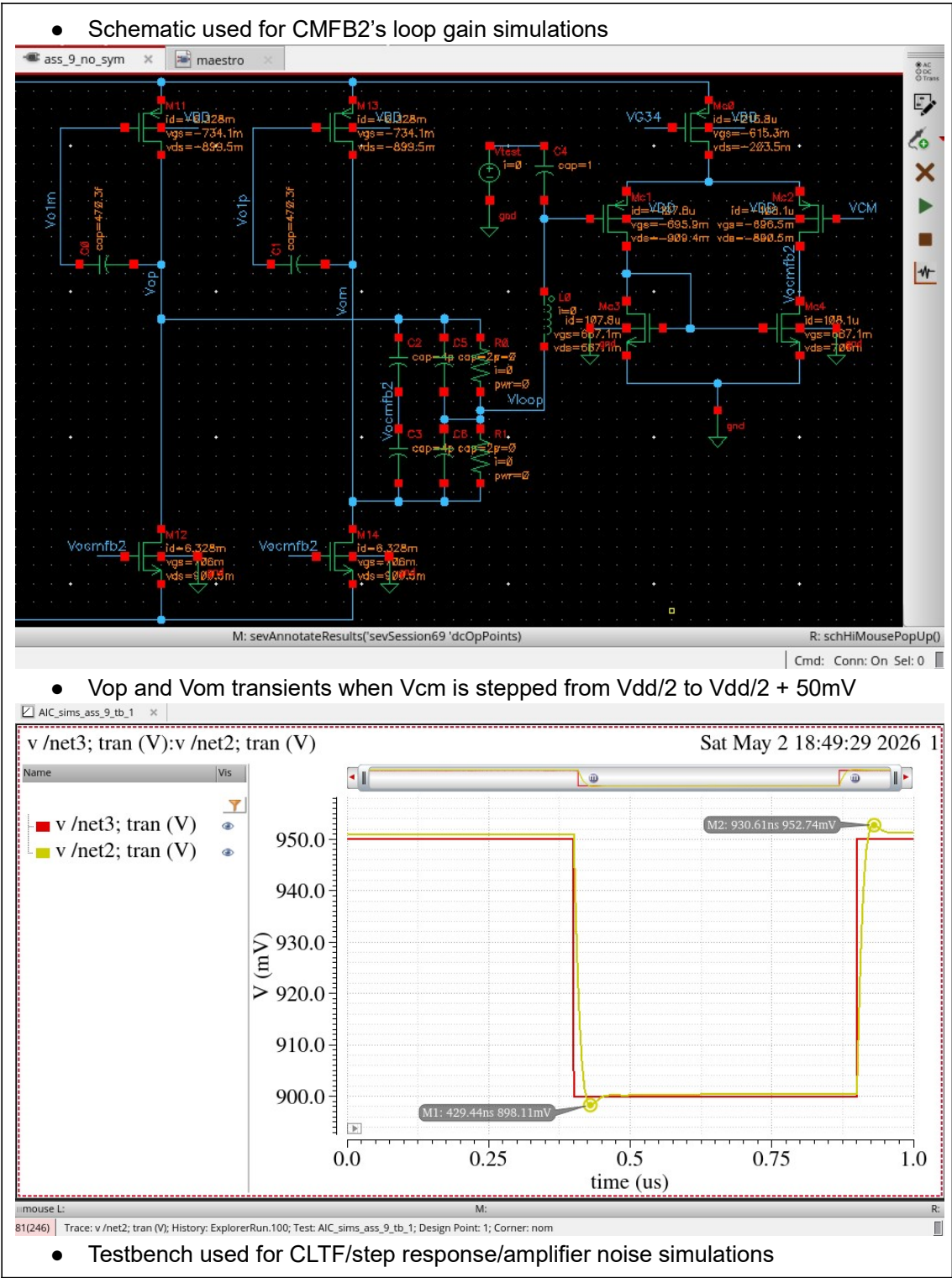


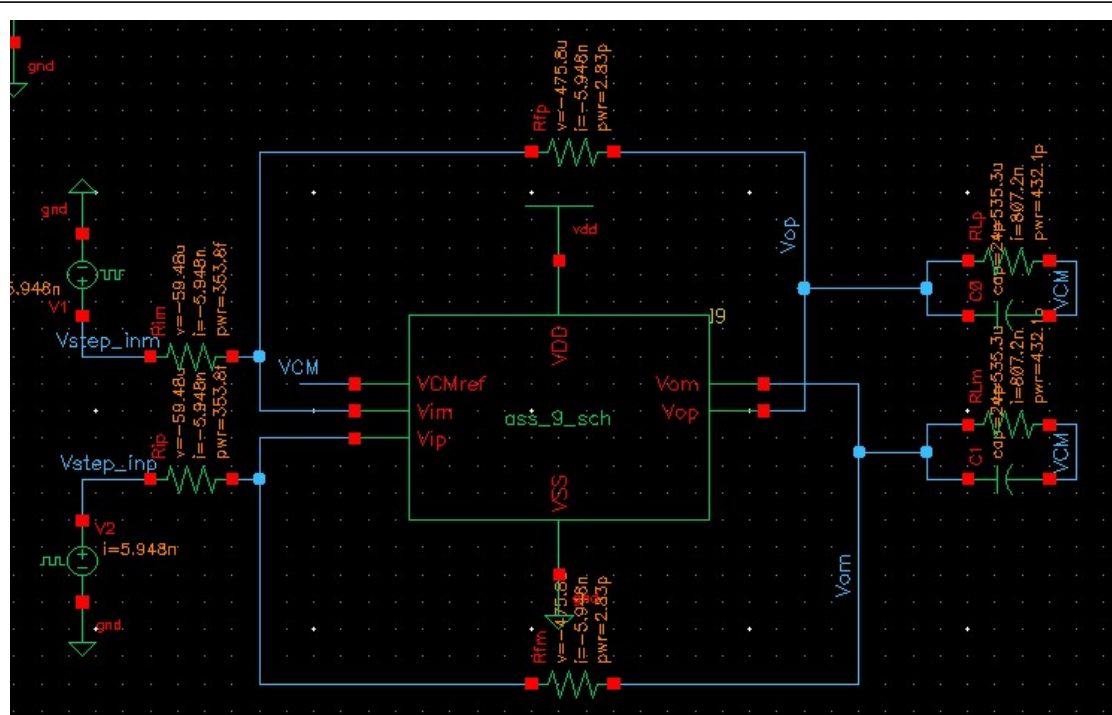
- Schematic used for the opamp's 2nd stage



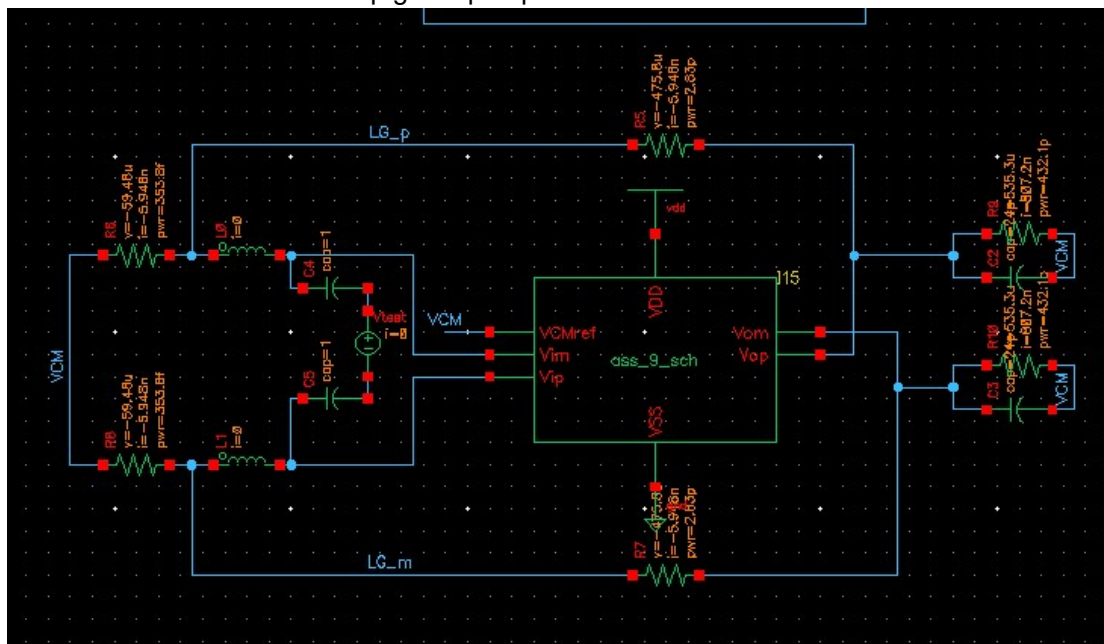
- Testbench used for DC simulations and CMFB2 transients







- Testbench used for loop gain/opamp noise simulations



- Testbench used for HD3 simulations

